

Supplement: Isolon Exclusivity Proof Surface

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This optional file lists named lemmas in the host-only Verus crate `ept_model` (2,366 lines). The conference paper is self-contained without it. Live executable EPT is not in this list and is not L3.

1 How to replay

Pin Verus release/0.2026.07.12.0b42f4c (Linux zip SHA-256 in `verus-version.toml`). Then `cargo verus verify -p ept_model`. Expected: 80 verified, 0 errors, including the N -guest CI alias. No admit on the exclusivity path. The theorem is guest-exclusivity, not hypervisor non-access.

2 Named theorems

Identifier	Claim
<code>theorem_single_guest_4k_map_unmap_exclusive</code>	finite enabled 4K sequences preserve exclusive ownership
<code>theorem_n_guest_4k_map_unmap_exclusive</code>	same discharge; N -guest CI name
<code>theorem_concrete_single_guest_4k_refine</code>	scoped $\text{ghost} \leftrightarrow \text{exec}$ commute (one guest)
<code>theorem_concrete_n_guest_4k_refine</code>	scoped refine for N guests
<code>theorem_large_page_map_unmap_exclusive</code>	2M/1G span map/unmap
<code>theorem_numa_map_unmap_affinity</code>	mapped frames recorded on ghost NUMA node
<code>theorem_hw_2m_leaf_refines_identity</code>	2M leaf encoding vs identity intent
<code>theorem_ept_violation_preserves_exclusive</code>	emulate / reject / claim-map
<code>theorem_page_transfer_preserves_exclusive</code>	unmap source, map same HPA at dest
<code>theorem_alloc_map_unmap_refines</code>	frame pool + EPT refine

3 Supporting lemmas (selected)

`lemma_empty_exclusive`, `lemma_map_ok_exclusive`, `lemma_unmap_ok`, `lemma_apply_step_exclusive`, `lemma_two_guests_map_distinct_fr`, `lemma_map_span_exclusive`, `lemma_unmap_span_exclusive`, `lemma_2m_map_unmap_exclusive`, `lemma_1g_map_unmap_exclusive`, `lemma_violation_noop_preserves_exclusive`, `lemma_violation_cl`, `lemma_transfer_preserves_exclusive`.

The crate contains further span, NUMA, and hardware-decode lemmas. Count them with the verifier, not this list: the CI number is 80 functions.

4 What this supplement does not prove

The UEFI binary, live memory/ept.rs, device emulation, HTTP, SPA, ISO install, and guest firmware. Nested distro sysinstall is not iron.